

No.

Date

5.1 4th 15th 16th の加減算

A-10 D-13

B-11 E-14

C-12 F-15

$$\begin{array}{r} 1001 \\ 0101 \\ \hline 1110 \end{array}$$

① 2進数

1) $1_2 + 0_2 = 1_2$

2) $1001_2 + 0101_2 = 1110_2$

3) $10011101_2 + 11011100_2 = 101111001_2$

4) $1_2 - 1_2 = 0_2$

5) $1010_2 - 0011_2 = 0111_2$

$$\begin{array}{r} 10011101 \\ + 11011100 \\ \hline 101111001 \end{array}$$

$$\begin{array}{r} 1010 \\ - 0011 \\ \hline 0111 \end{array}$$

② 16進数

1) $A_{16} + B_{16} = 15_{16}$

2) $1FA5_{16} + 11CB_{16} = 3170_{16}$

3) $C_{16} - 3_{16} = 9_{16}$

4) $FA10_{16} - ABCD_{16} = 4E43_{16}$

$$\begin{array}{r} 1FA5 \\ + 11CB \\ \hline 3170 \end{array}$$

$$\begin{array}{r} A \\ + B \\ \hline 15 \end{array}$$

$$\begin{array}{r} 2516 \\ 9016 \\ \hline EFA10 \\ - ABCD \\ \hline 4E43 \end{array}$$

$$\begin{array}{r} C \\ - 3 \\ \hline 9 \end{array}$$

$$\begin{array}{r} 2516 \\ 9016 \\ \hline EFA10 \\ - ABCD \\ \hline 4E43 \end{array}$$

A-10 D-13
B-11 E-14
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5-2 補数の計算

① 2進数の2の補数

1) $1101_2 \Rightarrow 0011_2$

2) $01011001_2 \Rightarrow 10100111_2$

3) $110100011110_2 \Rightarrow 001011100010_2$

4) $01011100110_2 \Rightarrow 101000011010_2$

$$\begin{array}{r} 1111 \\ -1101 \\ \hline 0010 \\ +0011 \\ \hline 0011 \end{array}$$

$$\begin{array}{r} 11111111 \\ -01011001 \\ \hline 10100110 \\ +00000001 \\ \hline 10100111 \end{array}$$

$$\begin{array}{r} 110100011110 \\ +001011100001 \\ \hline 001011100010 \end{array}$$

$$\begin{array}{r} 01011100110 \\ 101000011001 \\ \hline 101000011010 \end{array}$$

② 16進数の補数

1) $89AB_{16} = 7655_{16}$

2) $FFFF_{16} = 0001_{16}$

3) $0AFD_{16} = F503_{16}$

4) $1234ABCD_{16} = EDCB5433_{16}$

$$\begin{array}{r} FFFF \\ -89AB \\ \hline 7654 \\ +1 \\ \hline 7655 \end{array}$$

$$\begin{array}{r} FF \\ -FFFF \\ \hline 0000 \\ +1 \\ \hline 0 \end{array}$$

$$\begin{array}{r} FFFF \\ -0AFD \\ \hline F502 \\ +1 \\ \hline F503 \end{array}$$

$$\begin{array}{r} FFFF FFFF \\ -1234 ABCD \\ \hline EDCB 5432 \\ +1 \\ \hline EDCB 5433 \end{array}$$

5-3 補数計算

1) $0110_2 - 0010_2$

$0110_2 + 1110_2 = 10100_2 = 0100_2$

$$\begin{array}{r} 0110 \\ - 0010 \\ \hline 0100 \end{array}$$

$$\begin{array}{r} 0101 \\ - 0110 \\ \hline 10100 \end{array}$$

$$\begin{array}{r} 1010 \\ + 1110 \\ \hline 11000 \end{array}$$

2) $1010_2 - 0010_2$

$1010_2 + 1110_2 = 10000_2 = 1000_2$

3) $1011_2 - 1010_2$

$1011_2 + 0110_2 = 10001_2 = 0001_2$

$$\begin{array}{r} 1011 \\ + 0110 \\ \hline 10001 \end{array}$$

$$\begin{array}{r} 01100101 \\ + 10101101 \\ \hline 100010010 \end{array}$$

4) $01100101_2 - 01010011_2$

$01100101_2 + 10101101_2 = 100010010_2 = 00010010_2$

5) $01001100_2 - 11010011_2$

$01001100_2 + 00101101_2 = 01111001_2$

$$\begin{array}{r} 01001100 \\ + 00101101 \\ \hline 01111001 \end{array}$$

6) $11001100_2 - 01010011_2$

$11001100_2 + 10101101_2 = 101111001_2 = 01111001_2$

$$\begin{array}{r} 11001100 \\ + 10101101 \\ \hline 101111001 \end{array}$$

7) $11011100_2 - 11010001_2$

$11011100_2 + 00101111_2 = 100001011_2 = 00001011_2$

$$\begin{array}{r} 11011100 \\ + 00101111 \\ \hline 100001011 \end{array}$$

5-4 10進数を符号化

① BCD 符号

$$1) 1234_{10} = 0001001000110100_{BCD}$$

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0001 & | & 0010 & | & 0011 & | & 0100 \end{array}$$

$$2) 9876_{10} = 1001100001110110_{BCD}$$

$$\begin{array}{cccc} 9 & 8 & 7 & 6 \\ 1001 & 1000 & 0111 & 0110 \end{array}$$

② グレイ符号

10進数	グレイ符号
0	0 0 0 0
1	0 0 0 1
2	0 0 1 1
3	0 0 1 0
4	0 1 1 0
5	0 1 1 1
6	0 1 0 1
7	0 1 0 0

10進数	グレイ符号
8	1 1 0 0
9	1 1 0 1
10	1 1 1 1
11	1 1 1 0
12	1 0 1 0
13	1 0 1 1
14	1 0 0 1
15	1 0 0 0